

A Pre-feasibility Study for the Establishment of Granite Extraction Plant



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1. Executive summary

The granite stone is one of the raw material occurred abundantly in Jordan, the establishing of a plant for extracting the ore is economically feasible idea. One of the most abundant areas in Jordan for this ore is Aqaba which is located in the southern part of Jordan city. Granite stone is important in the construction process.

based on the market share and prices, (which will be presented in deatails later on), which were considered for the project, the estimated revenues for the first year of production will be about 3,546,881 JD, gradually increased to reach 4,627,876 at the 10 th year of the project, the payback period is 3 years and the internal rate of return is 35.19%, in addition to generate 13 job opportunity to the governorate residents.

Table1 : Project Main Indicators

Results	
Total Investment cost (JD)	1,642,154
Internal Rate of Return	%35.19
Payback Period (Years)	3
Net Present Value (JD)	3,630,519
Job opportunities	14
Estimated Annual Revenues for 10 years (JD)	40,661,019
Total Operating Cost for 10 years (JD)	32,874,434
Gross Profit for 10 years (JD)	3,272,836

2. Key Highlight of Jordan

Jordan Hashemite Kingdom area is 89342 Kilometers with a population exceed 9.8 Millions distributed among 12 Governorates.

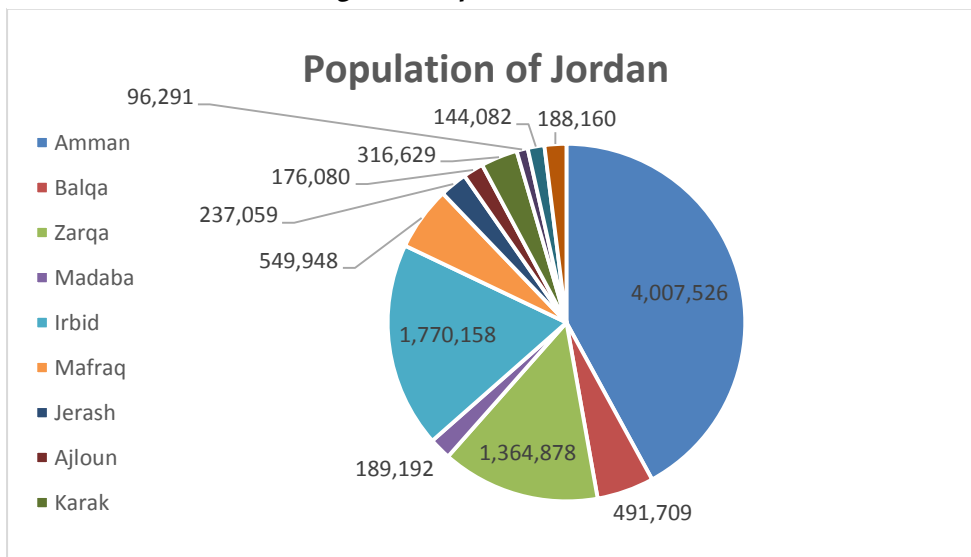
Jordan is bounded by Saudi Arabia, Iraq, Syria and Palestine from the South, East, North and west respectively. Jordan climate is mostly arid desert with rainy season between November to April Months.

Jordan is a free market economy, ranked as the fifth freest economy in the MENA region in the Index of Economic Freedom. Its free market economy enjoys strong partnerships amongst its neighboring country as well as Europe and the USA. Jordan is a signatory of several bilateral and multilateral trade agreements such as (Free Trade Agreement with the US and the EU, a Free Trade Agreement with the EFTA states, an FTA with Singapore, A member of the Greater Arab Free Trade Agreement (GAFTA) and a signatory of the Aghadir Agreement. Looking at the range of countries these agreements cover and facilitate trade between, one is confident to say that Jordan has business gateways and partnerships across the globe.

Population Overview

Based on the latest surveys done by Department of Statistics (DOS) in 2017, total number of population has reached around 9.814.995 with an increase rate of 2.88% from 2016 year, figure below summarizes the population distribution among the kingdom governorates.

Figure 1: Population of Jordan



Department of Statistics (DOS), 2015

Moreover, and based on the latest survey conducted by (DOS) which related to population nationality distribution over the governorates, it was found that total Non-Jordanian residents constitutes around 30% of the total population. Half of these are Syrians (1.3 Million) concentrated mainly in Amman Capital (436 Thousand) then Irbid (343 Thousand), Mafraq at 208 Thousand and Zarqa at 175 Thousand.

Jordan prides itself in its youthful population with 35% of ages below 15 years old and only 4% with ages above 65 years old. Table below summarizes population distribution of Jordanians according to Age Group:

Table2 : Population Distribution according to Age Group

Age Group	Population %	Males	Females
0-14 Years	%35.04	%19.2	%18.2
15-64 Years	%61.02	%30.7	%28.6
More than 65 Years	%3.94	%1.6	%1.6

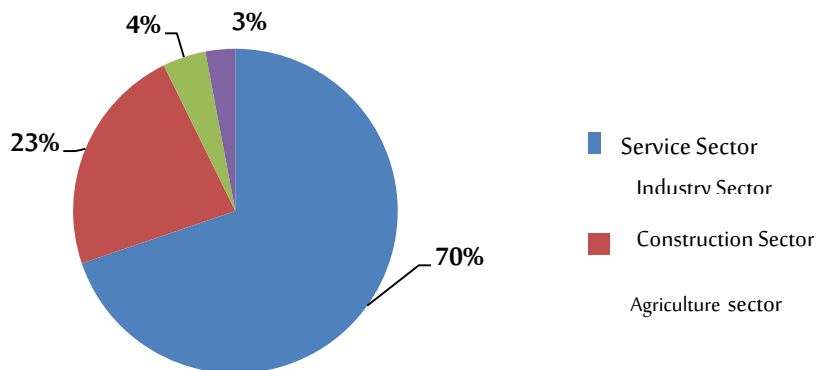
Department of Statistics (DOS), 2015.

Overview of Jordan Economy

Classified as an upper middle economy by the World Bank and as a country of High Human Development by the UNDP, Jordan is an important financial power in the Middle East. Its small market witnessed growth in the last two decades since King Abdullah II accessed the throne. Jordan's GNI per capita has increased in the last four years due to the basket of economic reforms that were enacted in the recent decade such as (the new Income Tax Law, Public Private Partnership Law and Investment Law). These laws aim to encourage the participation of the private sector in the Kingdom's economic development and provide an enabling legislative environment for current and new investment opportunities.

The Jordanian economy is overwhelmingly services oriented and its contribution in the GDP is 68.1%. The contribution of the industrial sector is 22.4%, the construction sector is 4.2% and the agricultural sector is 2.9%. These contributions are aimed to increase to (27.4%), (5.8%) and (3.4%) respectively and that of the service sector is due to decrease according to Jordan 2025 vision.

Figure 2: Sectors Contribution to Jordan's Economy



Jordan's exports include variety of textiles, potassium, phosphates, fertilizers, vegetables and pharmaceutical products. While it's main imports are crude and refined petroleum. Distinguished by its strategic location, on the crossroads between Asia, Africa and Europe with strong connections to the Levant and the GCC, Jordan has a regional market of interest that represents US\$3.8 trillion market and comprising 380 million consumers.

Jordan infrastructure ranks comparatively well (38th out of 148 comparable economies), with an extensive 8000 KM road network connecting Jordan domestically and externally The new Queen Alia International Airport and the Port of Aqaba are the major gateways to the international market. In addition to some mega projects such as the Red- Dead Sea Canal and the national railway network that will be developed to position Jordan as a hub for regional commerce.

Jordan banking system is quite sophisticated, resilient and in compliance with international standards, making it very attractive and trustworthy to investors. This is reflected in the fact that 50% of equity in licensed banks in Jordan is held by non-Jordanians, and non residents' deposits in Jordanian banks witnessed a steady growth of 19.2%.

Moreover, realizing the value of MSMEs to drive economic growth, the government has developed the national Strategy for the encouragement of entrepreneurship and the development of micro small and medium sized enterprises for 2015-2019. This shows the commitment of the Jordan government to enhance the private sector development and leveraging the country's strong human capital.

Jordan prides itself in its youthful population. It's the country most valuable capital. A tech savvy well educated and trained workforce attracts a lot of investors to Jordan. More than 20.4% of Jordan's GDP is dedicated to the education and capacity building for the labor force, this has resulted in securing a 91% literacy rate and enabled Jordanians to be among most hired and qualified middle-eastern workforce. Such solid, diverse and resilient characteristics of the Jordanian economy and investment scene position Jordan as a competitive investment destination.

Major Economic Indicators

- **GDP Growth**

The Jordanian economy slowed and reached 2.4 percent in 2015 compared to growth rate of 3.1 in 2014 and similar to MENA growth rates.

The growth also slowed for trade, restaurants and hotels; manufacturing; transport; and electricity and water. Meanwhile, finance, insurance, and business services advanced at a faster pace.

- **Inflation rate**

Inflation rate decreased to 0.9 in 2015 after 2% decline in the previous year. Inflation Rate in Jordan averaged 3.1 percent from 2011 until 2015.

- **Unemployment**

The unemployment rate in Jordan was recorded at 13 percent in 2015, comparing to 11.9 percent a year earlier. This increase due to the current instability in the labor market structure as well as high competition from low cost foreign labors especially from Syria.

- **External Sector**

A number of risks manifested in 2015, the closure of land trade routes with Syria and Iraq and the deepening instability in the region adversely impacted many external sector indicators such as trade, tourism and direct Investment.

Moreover, loans disbursements increased by JD 545.3 million, due to the use of the International and Arab Monetary Funds (IMF and AMF) credit facilities. As an outcome of these developments, the overall balance of the balance of payments registered a surplus of JD 328.7 million in 2015, compared with a surplus of JD 1,550.7 million in 2014.

Further, net international investment position (IIP) witnessed an increase in the Kingdom's net obligations to abroad; to reach JD 24,357.5 million, compared to a net obligation of JD 22,578.8 million at the end of 2014 as a result of the increase in the stock of external financial assets and liabilities of all resident economic sectors to reach JD 18,657.9 million and JD 43,015.5 million; respectively.

External Debt in Jordan increased to 9390.50 JOD Million in 2015 from 8030.10 JOD Million in 2014. External Debt in Jordan averaged 5433.67 JOD Million from 1988 until 2015, reaching an all time high of 9390.50 JOD Million in 2015 and a record low of 3640.20 JOD Million in 2008.

Investment Climate in Jordan

Investment in Jordan is considered as one of the vital sources to boost the economy considering that Jordan's economy is among the smallest in the Middle East, with insufficient supplies of water, oil, and other natural resources, underlying the government's heavy reliance on foreign assistance.

The country's location, supported by myriad free trade agreements (FTAs) offering access to 1.5bn consumers, enables the kingdom to be a strategic trade route to many of its neighboring countries and regions.

Jordan aspires to create a competitive investment destination capitalizing on its many advantages mentioned above. The government focused its efforts to implement significant advances in structural and legal reform. These efforts are represented in the new Investment Law of 2014, the tax law, in addition to other endeavors related to providing greater access to credit for MSMEs, and by providing greater investment incentives to specific priority sectors identified by the new Investment law and the Jordan 2025 vision.

Furthermore, in its endeavor to enhance the business environment, several geographically industrial states were created across the kingdom. These range in type to include (industrial estates, free zones and special economic zones). Under the new Investment Law, these zones are given a number of incentives that include a tax rate of 0% on exports, sales tax, import duties, social service tax, dividends tax and a 5% income tax from all economic and manufacturing activities undertaken in the development zones. As for the investments in Free Zones, they enjoy Exemptions from customs duties, income tax exemptions and exemptions from land and building taxes among others. Both development and free zones also enjoy facilitations regarding visa and residency permits for investors and workers in addition to Repatriation of capital and profits in a convertible currency. Such estates have succeeded in attracting relatively large amounts of FDI to Jordan.

Key National Investment Priorities:

Jordan Investment commission worked on taking a leading role in the application of government policies to promote and attract domestic and foreign investments and create an investment environment that stimulates economic performance through investment promotion strategy launched in 2016

Jordan investment commission aims to:

- Regulating the special provisions governing Development Zones and Free Zones in the Kingdom and developing them placing them in service of the national economy as well as monitoring their functioning.
- Developing plan and programs to stimulate domestic and foreign investments.
- Establishing trade centers and organizing exhibitions as well as opening markets and organizing trade missions in order to promote national products, in addition to marketing and development of national exports and encouraging investment.
- Taking appropriate decisions related to private or public institutions to improve Investors' confidence in Jordan's investment environment.

3. Market analysis

Project description

Due to the multiple uses of granite ores, it is feasible to set up a quarry to extract this raw ore to be marketed after cutting and polishing, which will create benefits to the local community in terms of employed of labor and supplying the local market with high-quality granite products as well as increasing exports volumes to outside countries.

Granite

An igneous rock formed under high temperatures. Granite is formed mainly of three minerals quartz and aluminum silicate alkali and plagioclase. These minerals give the granite different colours as white or crimson or gray. Granite also contains small amounts of minerals dark brown or dark green or black, such as mica and hornblende biotite. The granite granules is characterized as big size, so its easily, to find out the difference. The width of many atoms is of 0.5 cm. There are minerals in the granite interact. As a result, it is considered a strong endurance granite rock solid and useful in the construction of buildings.

Most of the granite can resists erosion for centuries, and it is able to be polished to becomes smooth, making it viable for the construction of the columns, tombstones and monuments, which will be lasted for a long time. There are other types characterized by granite in terms of tissue, such as albraufabra fabric which is characterized the granite from the rest of igneous rocks and this tissue indicates that the granite freezes on the first two phases slowly and then rapidly creating Braufabra fabric mixture of clear and accurate crystals. Origin characterized by fabric coarse-grained because of slow cooling below the surface, allowing the growth of crystals and clarity.

Geologists classify granite as igneous rock. They have concluded that most of the granite be mediated crystallization of magma (melted rock material) Slow cooling. Magma and its chemical composition of the same granite, and resulted from the fusion of rock, ranging between 25 and 40 km beneath the surface under land, in a temperature ranges between 650 and 9000 ° m. Magma rises because it is lighter than the surrounding solid rock. And cooled magma higher. And mostly cool slowly enough to form coarse crystals harden beneath the surface of the earth. The magma seeps sometimes volcanoes, and rapidly cooled to be large crystals, known as the resulting Rock of Balriuliet or lava granite, and has the same granite chemical composition, but with soft atoms. Experience has shown that many types of rocks produces granite when fused.

The rocks fused on stages, but the minerals which formed the granite fused first. It may be the reasons for the existence of granite abundantly ease of melting rocky article. Continents consists largely of granite buried under the sedimentary rocks. Most of the granite appears when buried deep rocks rising to the surface as a result of movements of mountain formation on the Earth's crust. When the mountain peaks out cropped as a result of erosion the granite rocks appears underneath.

Specifications

Chemically classified as igneous rock acidic because the specific gravity is low and of light colour which indicates that the proportion of silicate minerals is more than 65%, such as metal and Alblajoclaiz quartz and biotite and Almskovit.

Forms

Different forms, colours, and surface sizes, some customers prefer small sizes granite or large-scale measurments which is more beautiful when being installed. Thickness of shearing also varies based on decoration requirments and measurments as well as thickness; which is usually thick for floor uses and thin for walls, in addition to variation of the surface which is either polished or matt.

Types of granite

Industrial Granite: Granite is a stone with a high cost when extracted and polished; where its relying on labour during the extraction and sharing processes in a form of plates which is highly cost, as a result it appeared the so-called industrial granite or marble. Industrial marbles from breaking or cutting marble, cement and chemicals to be manufactured in the form of panels composed, characterized to be free of cracks or holes, and the multiplicity of colours and decorations and hardness and resistance to factors of friction, usually panels divided into less thickness than natural granite. Industrial marble used in broader aspects than the natural marble; where it was easy to use in stairs and the building faces, as well as in flooring and other uses.

Natural granite: Many types of granite spreads in earth, which is differ in hardness and the proportion of minerals, which govern the colour and hardness, as a result of the slow chill of volcanic magma which formed the granite. There are many types of natural granite, which can be distinguished by the area where the granite occures.

Lithology of Granite:

Observation by the naked eye: through observation of the naked eye or manual Magnifier or magnifier double clarified that granite is composed of crystals in forms of granules which are recognized

By shape and color:

Crystals Feldspat (Feldspath): with a regular shape (usually rectangular) and different colours

Crystals mica (Mica): in the form of shiny leaflets minutes and shiny, exist either in black or white.

Quartz (Quartz Crystals) gray likes glass

Granite was formed by crystallization of magma (melted rock material) by slow cooling. Magma has the same chemical composition of the granite, and resulted from the fusion of rocks for a distance ranging between (25 to 40 kilometers) beneath the ground, in a temperature ranges between (650 and 900 ° m). Magma rises because it is lighter than the surrounding solid rock. The magma will get cooler when it rises to reach the surface. And mostly cool slowly enough to form coarse crystals harden beneath the surface of the earth. The magma seeps sometimes volcanoes, and rapidly cooled to form large crystals, the resulted rocks are known as Balriuliet or lava granite, and has a chemical composition of the same granite, but the crystals are smoother. Experience has shown that many types of rocks produces granite when fused. Continents consists largely of granite buried under the sedimentary rocks. Most of the granite appears when buried deep rocks rising to the surface as a result of movements of mountain formation on the Earth's crust. When peak of mountains out cripes as a result of erosion, the granite rocks underneath, appears.

The most important characteristics of the magma are:

- 1 - Magma characterized consists mostly of silica.
- 2 - Magma characterized by high temperature.
- 3 - Magma characterized by its ability to movement and its ability to flow and emergence. This is true despite the fact that some magma almost been solid.

Table 3: Physical properties of granite ore

Properties	
colour	Gray with black points
Surface Properties	rough
fracture	angular
Homogenization	Non-homogenous, composed of three elements: Quartz: (al maro) transparent and gray Feldespar: white or roasy Mica: white/ black
Permeability	No permeability
Acid Influnces	No influences

Table 4: chemical composition

Composition	Percentage
SiO ₂	% 73,86
Iron Oxides Fe ₂ O ₃ , FeO	% 1.91
Magnesium Oxide MgO	% 0.26

Granite: Acidic rock because it contains a proportion of Sils: 73.86% , higher than 66%, and light colour because of poor iron and Amonez $(1.91 \ 0.26) = 2.17\%$

Project Objectives

- Exploitation of millions of tons of granite ores, to be used in buildings and decoration industries.
- Exploiting of granite ores leads to obtain highly pure products encouraging to use these products in the local industries to replace foreign imports, resulted of saving foreign currency.
- enhancing the development affairs by the exploitation of domestic mineral resources to be used in many industries, currently decoration works, decoration and buildings.

proposed product

Uses:

This type of rock is widely used to carve statues and columns, which is characterized by being un effected of corrosion and erosion more than the sedimentary rocks. The granite got a great economic importance and is also used in homes, kitchens and stairs, well known the Saudi granite.

Granite in flooring:

Granite is used inflooring, spesifically in flooring villas and palaces where it gives the latter an aesthetic and serene view of a luxury of the place, but shall be drawn to the fact that the marble is not used for all floors for example, it is not recommended for use in flooring that she has to absorb the tremendous pressure of the weights warehouses and laboratories

Granite in the exterior of the buildings:

The granite is used abundantly in buildings facing , but the use of granite in the exterior of the buildings is different in terms of texture where granite is tough texture, not smooth is used, the weight and space of granite during the planning should be taken into account.

Granite for stairs and bottom walls:

The use of granite in the stairs and the bottom of the walls or the so-called Balseklo (skirting), and imparts an aesthetic and creative touch, especially if the wall in turn is granite.

Granite in the bathrooms and laundries:

Granite is used to decorate the cabins in the bathrooms and offer a variety of forms and colours for this purpose, and shall refer to the necessity of granite thickness of 3 cm for sinks, sinks should be chosen in advance to match cut with its requirements.

Granite counters top and shelves Cuisines:

The granite is used as counter top for the kitchen since its solid and bear works of the kitchen, as well as it is used in racks, it is advised to choose the thickness of 3 cm for this purpose with a choice of dark colours instead of light colours and being oiled by insulating material, and polished the granite and rounded edges

Other uses:

- Coverages of doors and windows frames.
- Art designs, sculptures and pieces of art using granite as a feedstock.
- The use of granite in the formation of the mosaic.

Target Segments

- contracting companies
- kitchen manufactures companies
- Decoration companies

Market Size Analysis

There are about 2000 plant for the production of marble, granite and stone in Jordan.

The Estimated total investment for marble and granite sectors, according to Stone & Marble Association, is about half a billion jordanian dinars, the equivalent of \$ 700 million, the export volume last year was about \$ 30 million based on the association statistics.

Competitors Analysis

- Radwan Granite factory, which is characterized by having all the machines which are required to import of irregular blocks a block irregular marble or granite, to be cut and produced accurately according to customers demand.

- Khalifa Granite factory
- Melhem Granite factory
- saleem Al-zmar Group for Marble and Granite
- First Quarries Company for Granite
- others

Price analysis and pricing policy

Granite prices used in decoration works in the domestic market ranging between 7-30 JD as product classification, based on the exports and imports quantities of granite ores there is a great chance to compete with production capacity estimated at 5,000 m³/year in addition to export. The table below shows estimated annual revenues of the project

Table 5: estimated annual revenue of the project

Products	Percentage	The annual production (m ³)	Price m ² (JD)	Price m ³ (JD)	Annual revenue (JD)
Main product (polished granite)	85%	4,250	30	-----	4,249,995
Secondary product (broken granite)	15%	750	-----	7	5,250
Total	100	5,000	-----	-----	4,255,245

Market Share Analysis

A quarry of 5,000 m³ production capacity will be established to compete at 0.25% market share based on technical experience, which can be increased in the future considering the supply and demand in domestic market.

Table 6: Revenues of the project during the years 2018-2027

Prices and revenues assumptions	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Polished granite price per sqm	25	26	27	27	28	29	30	31	32	33
Cracked granite price per cubic m	7	7	7	8	8	8	8	9	9	9
Polished granite revenues	3541631	3647880	3757317	3870036	3986137	4105721	4228893	4355760	4486433	4621025
Cracked granite revenues	5250	5408	5570	5737	5909	6086	6269	6457	6651	6850
Total Revenues	354688	365328	376288	387577	399204	411180	423516	436221	449308	462787
	1	8	6	3	6	7	2	7	3	6

4. Technical analysis

This section is clarifying the technical specifications of the project in addition to estimate and review the most important items of capital and operating costs based on the market study results. The technical analysis will explain and generally describe the project components as well as the construction and operating requirements of the project, the general specifications where the cost was based on will be identified based on the estimations of technical team of the study

Required Human Resources

Table 7: Required Manpower

Job Title	No.
Project manager	1
Technical cutter	2
Accountant	1
Labour	2
driver	4
Security guard	2
Cleaner	1
Total No. of Employees	13

The following shows the job description for the required human resources:

- Project Manager: works independently according to the given responsibility, proposing the policies and work plans, representing the management planning and managing the work at the plant. follow up the execution of plans and programmes in compliance with decided policies, objectives and targets of the plant in a proper way to achieve the owner targets.
- Technical cutter: carry out the cutting of the blocks in a technical method after being extracted from the quarry.
- Accountant: in charge of plant cost expenses in addition to all related financial and auditing aspects of the plant and preparing the annual budget.
- labours: carried out the drilling and extraction works at the quarry.
- Drivers: transpending the granite ore from the quarry to the plant to be manufactured and processed.

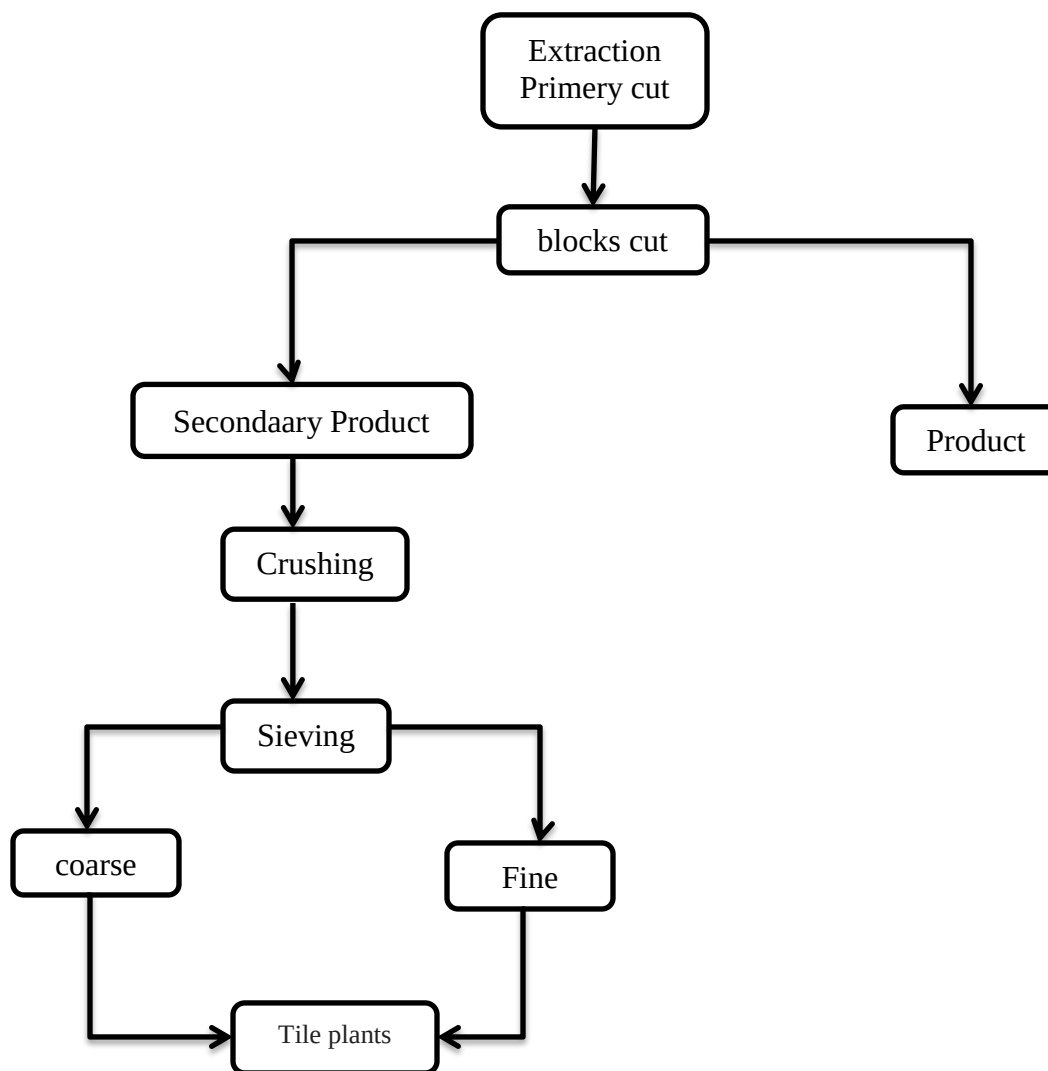
- Security Guard: excluding the approved security procedures to keep the quarry secure and hygiene.
- Cleaners: carry out the cleaning works to keep the offices, the plant, toilets and all facilities clean.

Site Analysis

The plant will be the establishment of the Aqaba Special Economic area where it will be taking the right of exploration and then mining in the southern area of Aqaba.

Description of Production Process

Figure3 Production processes



based on the progress scheme of operations as described above, the ore granite will be extracted from occurrences sites, carry out the primary cut to get blocks, then cut the blocks to get panels and finally polished to be used for decoration, the secondary rubble become valid uses of decoration, it will produce a secondary rubble products will be crushed by jaw crusher and then sieved to be used in the manufacturing of tiles.

Analysis of technical requirements

Land and construction works

The plant will be located in the southern governorate of Aqaba, where a 50,000 square meter land will be leased.

Table8 : Required Land

Land	Area (sqm)
The Required Land	50,000

The required construction works for the project will be as follows:

Table 9: Required construction works

constructions works	Area (sqm)
Building works	150
Hanger	500

Machinery and Equipment

The table below shows the equipments and machines required for the project:

Table 10: The Required Machinery and Equipment

Machinery and Equipment	Qty
Loader	1
Crushing machine	1
Dumper	3
Tipper Truck	1
Sieving machine	1

Furniture

Offices furniture includes (computers, tables and chairs, carpet and other requirements) with total cost of 10,000 JD.

Vehicles and Transportation

The table below shows the transportation and vehicles:

Table 11 required vehicles

Vehicles and transportation	Qty
Pickup Truck	2

Requirments and legal procedures

The following figure shows the necessary procedures for exploration and mining rights license depending on body regulating energy and minerals sectors:

Figure 4 The legal procedures-Source: Organization of Energy and Mines Commission (TRC).



5. Financial Analysis

The Financial Part illustrates all the assumptions which were used when we have developed the study including project expenses related to capital expenditures, operating expenses (Fixed and Variable costs) followed by illustrating project revenues.

Assumptions

- The weighted average cost of capital used in the study is 10.75% as it was calculated considering risk free and debt Interest rates as well as Market risk premiums to consider alternative opportunity costs for investor.
- Project working capital has been estimated and added to project cash outflows at the beginning of the project and annual increase in working capital has been also estimated and added to the operating expenses then net working capital has been added to total cash inflows at the end of project lifetime.

The time span for the financial study is 10 years starting from 2018, where annual increase rates have been estimated as per the tables below:

Assumptions

Table 12: General Assumption

General Information	
All Financial Numbers (Currency)	Jordan Dinars
Financial Study Time Span (Years)	10
Expected Project First Operating Year	2018
Last Year in the Financial Study	2027

Table 13: Currency Exchange Rates

Exchange Rates	Jordan Dinars
American Dollar	0.708
Euro	0.760

Table 14: Annual Growth Rates Assumptions

	Average
Annual Population Growth Rate	2.20%
Annual Sales Price Increase Rate	3.00%
Annual Expenses Increase Rate	3.00%
Annual Increase Rate in Employees Salaries	4.1%

(1) and (2) Source: Inflation Rates, Central Bank of Jordan

(3) Source: Social Security

Table 15: Expenses Assumptions

Expenses Assumptions	
Raw Materials and Packaging from Total Revenues	65.00%
Utilities Expenses from Total Revenues	5.0%
Maintenance Expenses from Total Revenues	2.50%
Depreciation Expenses from Total Revenues	1.00%
Insurance Expenses from Total Assets Value	0.75%
Marketing Expenses from Total Revenues	0.25%

Table 16: Income Tax Assumptions

Income Tax Assumptions	Value
(1) Average Income Tax in Jordan	20%
Natural wealth extraction (JD / ton)	0.50
Income Tax after Deduction	20%
(3) Compulsary Reserve Percentage	10%
Other Assumptions	Value
(1) Annual Monthly Salaries have been calculated after multiplying monthly salaries by:	16

Table 17: Risk Premium

Risk Premium	
(1) Risk Free Rate of Return	6.50%
(2) Return to Debt Maturity	9.48%
(3) Market Risk Premium	10.12%
Income Tax Rate	20%
(4) Beta	0.89
Growth Rate	4.00%

(1) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(2) Source :Central Bank of Jordan

(3) Source :Damodaran's Country Default Spreads and Risk Premiums Report

(4) Source :Damodoran Beta By Sector, , pages.stern.nyu.edu

Table 18: Weighted Average Cost of Capital

Weighted Average Cost of Capital	
Average Return on Risk Free Investment	%6.50
Return to Debt Maturity	%9.48
Market Risk Premium	%10.12
Income Tax Rate	%20
Bets	0.89
Equity	656,861
Loans Value	985,292

Loans	
Cost of Borrowing before Tax	%9.48
Cost of Borrowing After Tax	%7.58
Loans Value	985,292
Loans Percentage	%60
Equity	
Cost of Equity	%15.5
Equity Value	656,861
Equity Percentage	%40
Gross	
Project Value	1,642,154
Weighted Average Cost of Capital	%100

Capital Expenditures

The estimated capital costs of the project are 1,642,154 JD and cover the cost of land, construction, machinery and equipment, furniture and furnishings, vehicles and transportation, pre-operating expenses and initial working capital. These costs are shown in the following tables:

Table 19: Construction Work Capital Expenditure

Construction Works	Cost (JOD per Sqm)	Area (sqm)	Gross Cost
Hanger	100	500	50,000
Construction Works	200	150	30,000
Other			8,000
Gross Cost			88,000

Table 20: Machinery and equipment Capital Expenditure

Machinery and equipment	Quantity	Unit cost (JD)	Gross Cost (JD)
Loader	1	60,000	60,000
Crushing machine	1	40,000	40,000
Saw for cutting	3	25,000	75,000
Dumper	1	400,000	400,000
Sieving machine	1	25,000	25,000
Gross Total (JD)			600,000

Table 21 : Furnitutre Capital Expenditure

Furnitutre	Quantity	Unit cost	Gross Cost /Jordan Dinar
Office Furnitutre	-	-	10,000
Total			10,000

Table 22: Vehicle and Transportation Capital Expenditure

Vehicles and Transport	Quantity	Unit cost	Gross Cost /Jordan Dinar
Pickup Truck	2	23,000	46,000
Gross Total			46,000

Table 23: Pre-operating expenses

Pre-operating expenses	Gross Cost
Governmental fees	500
Transportation	750
Marketing	500
Training	750
Total	2,500

The initial working capital reflects the project's amounts needed to cover all the expenses during operation until the project starts generating income, the following table shows the components of the initial working capital:

Table 24: Initial working capital

Initial working capital	Number of months	Percentage	Cost (JD)
Operating expenses	3	%25	716,913
Administrative and general expenses	3	%25	4,397
Marketing expenses	3	%25	2,217
Indirect salaries	3	%25	22,840
Gross Total			746,367

Table 25: Summary of expenses

Summary of capital expenditures	Cost (JD)	Contingency	Cost (JD)	Percentage
Land	0	10%	0	%0.0
Construction Works	88,000	10%	96,800	%5.9
Machinery and Equipment	600,000	10%	660,000	%40.2
Furniture	10,000	10%	11,000	%0.7
Vehicles and Transportation	46,000	10%	50,600	%3.1
Pre-operating expenses	2,500	10%	2,750	%0.2
Initial working capital	746,367	10%	821,004	%50.0
Total (JD)	1,492,867		1,642,154	100%

Sources of funding

The financing structure required for the project consists of loans and equity. The investment cost of this project was estimated at 1,642,154 JD. This amount is determined through fixed assets, which include land, construction, machinery and equipment, furniture and furnishings, vehicles and transport vehicles, pre-operating expenses and initial working capital. (60%) of the project will be financed through loans (985,292 JD) and the rest (40%) through self financing (equity). The following table shows the general structure of the required financing:

Table 26: Sources and use of finance

Sources of funding	amount	Percentage
Equity (self-financing)	656,861	%40
Loans	985,292	%60
Gross Total	1,642,154	%100
Use of funding	amount	Percentage
Capital expenditure	818,400	%49.84
Pre-operating expenses	2,750	%0.17
Working capital	821,004	%50.00
Gross Total	1,642,154	%100.00

Operating Expenses

The following table shows the expected operating expenses of the project:

Table 27: Operating Expenses

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Raw material expenses	2,305,473	2,374,637	2,445,876	2,519,252	2,594,830	2,672,675	2,752,855	2,835,441	2,920,504	3,008,119
Service expenses	177,344	182,664	188,144	193,789	199,602	205,590	211,758	218,111	224,654	231,394
Maintenance expenses	88,672	91,332	94,072	96,894	99,801	102,795	105,879	109,055	112,327	115,697
Consumables	35,469	36,533	37,629	38,758	39,920	41,118	42,352	43,622	44,931	46,279
Other expenses	260,696	268,517	276,572	284,869	293,415	302,218	311,284	320,623	330,242	340,149
Gross Total	2,867,653	2,953,683	3,042,294	3,133,562	3,227,569	3,324,396	3,424,128	3,526,852	3,632,658	3,741,637

General and Administrative Expenses

The following table shows the expected administrative and general expenses of the project:

Table 28: administrative and general expenses

description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Communications expenses	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Hospitality expenses	600	618	637	656	675	696	716	738	760	783
Stationery and prints	600	618	637	656	675	696	716	738	760	783
Transportation and movements	1,200	1,236	1,273	1,311	1,351	1,391	1,433	1,476	1,520	1,566
Rent expenses	1,250	1,288	1,326	1,366	1,407	1,449	1,493	1,537	1,583	1,631
Training expenses	5,000	5,150	5,305	5,464	5,628	5,796	5,970	6,149	6,334	6,524
Insurance expenses	6,138	6,322	6,512	6,707	6,908	7,116	7,329	7,549	7,775	8,009
Other expenses	1,599	1,647	1,696	1,747	1,799	1,853	1,909	1,966	2,025	2,086
Gross Total	17,587	18,114	18,658	19,218	19,794	20,388	21,000	21,630	22,278	22,947

Marketing Expenses

Table 29: Marketing Expenses

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Marketing Expenses	8,867	9,133	9,407	9,689	9,980	10,280	10,588	10,906	11,233	11,570

Human Resources

The salaries and monthly wages of each employee were estimated to be multiplied by 16 months of the year for the purpose of calculating the increase in salaries and annual wages due to social security, health insurance, etc. to the employees of the administration. The following table shows the expected annual salaries and wages. The annual rate of increase in wages and salaries is taken into account:

Table 30: Expected monthly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project manager	1,500	1,562	1,626	1,692	1,762	1,834	1,909	1,987	2,069	2,154
Technical cutter	380	396	412	429	446	465	484	503	524	546
Accountant	400	416	433	451	470	489	509	530	552	574
Labour	300	312	325	338	352	367	382	397	414	431
Driver	400	416	433	451	470	489	509	530	552	574
Security guard	300	312	325	338	352	367	382	397	414	431
Cleaner	250	260	271	282	294	306	318	331	345	359

Table 31: Expected yearly salaries for the years (2018-2027)

Job Title	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Indirect Employees										
Project manager	24,000	24,984	26,008	27,075	28,185	29,340	30,543	31,796	33,099	34,456
Technical cutter	12,160	12,659	13,178	13,718	14,280	14,866	15,475	16,110	16,770	17,458
Accountant	6,400	6,662	6,936	7,220	7,516	7,824	8,145	8,479	8,826	9,188
Labour	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Driver	25,600	26,650	27,742	28,880	30,064	31,296	32,579	33,915	35,306	36,753
Security guard	9,600	9,994	10,403	10,830	11,274	11,736	12,217	12,718	13,240	13,782
Cleaner	4,000	4,164	4,335	4,512	4,697	4,890	5,091	5,299	5,517	5,743
Total indirect salaries	91,360	95,106	99,005	103,064	107,290	111,689	116,268	121,035	125,997	131,163

Depreciations

The following tables show all equipment costs, expenses and depreciation rates

Table 32: Capex and Depreciation Expenses

Costs of equipment, capital expenditures and annual depreciation	Cost (JOD)	Depreciation Rate	Annual Additions Percentage
Land	0	%0.0	%0.0
Construction Works	96,800	%5.0	%0.0
Machinaries	660,000	%10.0	%2.0
Furniture	11,000	%10.0	%5.0
Vehicles	50,600	%10.0	%0.0
	818,400		

Table 33: Depreciation and annual increases on construction work

Summary of capital costs, additions and annual	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Total fixed assets	818,400	832,150	846,192	860,531	875,176	890,132	905,408	921,011	936,947	953,226
Total consumption expenses	77,000	78,375	79,779	81,213	82,678	84,173	85,701	87,261	88,855	90,483
Total accumulated depreciation	77,000	155,375	235,154	316,367	399,045	483,218	568,919	656,180	745,035	835,517
Total additions	0	13,750	14,042	14,340	14,645	14,957	15,276	15,602	15,937	16,279
Total net book value	741,400	676,775	611,037	544,164	476,131	406,914	336,489	264,831	191,913	117,708

Loans

The following table represents the annual loan amount and the annual payment method with the details of the remaining amounts of loans for each year.

Table 34: Loan Details

Loan Value	985,292			
Annual Interest Rate	9.48%			
Loans Period	5			
Grace Period	0			
Loan Starts at year:	2018			
Annual Payment	256,474			
Number of Payments	5			

Year	Annual Payment	Interest	Equity capital	Loan Remaining Value
2018				985,292
2019	256,474	93,406	163,068	822,224
2020	256,474	77,947	178,527	643,697
2021	256,474	61,022	195,451	448,246
2022	256,474	42,494	213,980	234,265
2023	256,474	22,208	234,265	0

Income Statement

Table 35: Income Statement

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Revenues										
Sales Revenues	3,546,881	3,653,288	3,762,886	3,875,773	3,992,046	4,111,807	4,235,162	4,362,217	4,493,083	4,627,876
Gross Operating Revenues	3,546,881	3,653,288	3,762,886	3,875,773	3,992,046	4,111,807	4,235,162	4,362,217	4,493,083	4,627,876
Operating Expenses	-2,867,653	-2,953,683	-3,042,294	-3,133,562	-3,227,569	-3,324,396	-3,424,128	-3,526,852	-3,632,658	-3,741,637
Gross Operating Profit	679,228	699,605	720,593	742,211	764,477	787,411	811,033	835,364	860,425	886,238
Gross Profit Percentage	%19	%19	%19	%19	%19	%19	%19	%19	%19	%19
Salaries and Benefits (Indirect Staff)	-91,360	-95,106	-99,005	-103,064	-107,290	-111,689	-116,268	-121,035	-125,997	-131,163
General and Administrative Expenses	-17,587	-18,114	-18,658	-19,218	-19,794	-20,388	-21,000	-21,630	-22,278	-22,947
Marketing Expenses	-8,867	-9,133	-9,407	-9,689	-9,980	-10,280	-10,588	-10,906	-11,233	-11,570
Pre Operating Expenses	-2,750									
Gross Indirect Expenses	-120,564	-122,353	-127,070	-131,971	-137,064	-142,356	-147,856	-153,570	-159,509	-165,680
Income before Interest, Depreciation and Tax	558,664	577,251	593,523	610,239	627,413	645,055	663,178	681,794	700,917	720,558
Fixed Assets Depreciations	-77,000	-78,375	-79,779	-81,213	-82,678	-84,173	-85,701	-87,261	-88,855	-90,483
Income before Tax and Interests	481,664	498,876	513,743	529,026	544,735	560,882	577,477	594,533	612,062	630,076
Bank Interests	-93,406	-77,947	-61,022	-42,494	-22,208	-0	-0	-0	-0	-0

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Income Before Tax	388,258	420,929	452,721	486,532	522,527	560,882	577,477	594,533	612,062	630,076
Income Tax	-77,652	-84,186	-90,544	-97,306	-104,505	-112,176	-115,495	-118,907	-122,412	-126,015
Net Profit	310,606	336,743	362,177	389,226	418,021	448,705	461,982	475,627	489,650	504,061
Net Profit Percentage	%9	%9	%10	%10	%10	%11	%11	%11	%11	%11
Compulsory Reserves	-31,061	-33,674	-36,218	-38,923	-41,802	-44,871	-46,198	-47,563	-48,965	-50,406
Retained Earnings	279,546	582,615	908,574	1,258,877	1,635,097	2,038,931	2,454,715	2,882,779	3,323,464	3,777,118

Expected Cashflows Statement

The table below summarizes project cashflows statement, where net cash flows are positive during all years as below:

Table 36: Expected Cashflows Statement

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cash Inflows from Operating Activities										
Net Profit	310,606	336,743	362,177	389,226	418,021	448,705	461,982	475,627	489,650	504,061
Non-monetary adjustments										
Bank Interests	93,406	77,947	61,022	42,494	22,208	0	0	0	0	0
Depreciation	77,000	78,375	79,779	81,213	82,678	84,173	85,701	87,261	88,855	90,483
Total Operating Cashflows before Additions to Working Capital	481,012	493,065	502,978	512,933	522,907	532,879	547,683	562,888	578,504	594,543
Inventory (Increase/Decrease)	-192,123	-5,764	-5,937	-6,115	-6,298	-6,487	-6,682	-6,882	-7,089	-7,301
Accounts Receivables (Increase/Decrease)	0	0	0	0	0	0	0	0	0	0
Accounts Payables (Increase/Decrease)	93,697	2,811	2,895	2,982	3,072	3,164	3,259	3,356	3,457	3,561
Working Capital (Increase/Decrease)	-98,426	-2,953	-3,041	-3,133	-3,227	-3,323	-3,423	-3,526	-3,632	-3,740
Net Cashflows from Operating Activities	382,586	490,113	499,937	509,800	519,681	529,555	544,259	559,362	574,873	590,803

	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Cashflows from Investments Activities										
Fixed Assets (Procurement)	-818,400	-13,750	-14,042	-14,340	-14,645	-14,957	-15,276	-15,602	-15,937	-16,279
Net Cashflows from Investments Activities	-818,400	-13,750	-14,042	-14,340	-14,645	-14,957	-15,276	-15,602	-15,937	-16,279
Cashflows from Financing Activities										
Capital	656,861									
Loan Amortization	-163,068	-178,527	-195,451	-213,980	-234,265	0	0	0	0	0
Bank Interest Rate	-93,406	-77,947	-61,022	-42,494	-22,208	-0	-0	-0	-0	-0
Loans	985,292	0	0	0	0	0	0	0	0	0
Net Cashflows from Financing Activities	1,385,680	-256,474	-256,474	-256,474	-256,474	0	0	0	0	0
Net increase (decrease) in cash	949,866	219,889	229,422	238,987	248,562	514,599	528,984	543,760	558,936	574,524
Cashflows at the Beginning of Period	0	949,866	1,169,755	1,399,176	1,638,163	1,886,725	2,401,324	2,930,307	3,474,067	4,033,003
Cashflows at the end of Period	949,866	1,169,755	1,399,176	1,638,163	1,886,725	2,401,324	2,930,307	3,474,067	4,033,003	4,607,527

Expected Balance Sheet

The balance sheet is one of the main statements which the project depends on, as it reflects the financial position of the entity and is usually estimated on the last day of the financial year.

All financial information and analysis of the project shall be the estimated for the years (2018-2027) as shown in the table below:

Table 37: Expected Balance Sheet

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Assets										
Current assets										
Fund	949,866	1,169,755	1,399,176	1,638,163	1,886,725	2,401,324	2,930,307	3,474,067	4,033,003	4,607,527
Inventory	192,123	197,886	203,823	209,938	216,236	222,723	229,405	236,287	243,375	250,677
Accounts receivable	0	0	0	0	0	0	0	0	0	0
Total assets traded	1,141,989	1,367,641	1,602,999	1,848,101	2,102,961	2,624,047	3,159,712	3,710,354	4,276,378	4,858,204
Non-current assets										
Fixed assets (net)	741,400	676,775	611,037	544,164	476,131	406,914	336,489	264,831	191,913	117,708
Total non-current assets	741,400	676,775	611,037	544,164	476,131	406,914	336,489	264,831	191,913	117,708
Total assets	1,883,389	2,044,416	2,214,037	2,392,265	2,579,092	3,030,961	3,496,201	3,975,184	4,468,291	4,975,912
Liabilities										
Accounts Payables	93,697	96,508	99,403	102,385	105,457	108,620	111,879	115,235	118,692	122,253
Remaining amount of Loan	178,527	195,451	213,980	234,265	0-	0-	0-	0-	0-	0
Total current liabilities	272,224	291,959	313,383	336,650	105,457	108,620	111,879	115,235	118,692	122,253

Description	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Non-current liabilities										
long term loans	643,697	448,246	234,265	0	0	0	0	0	0	0
Total long - term liabilities	643,697	448,246	234,265	0	0	0	0	0	0	0
Total liabilities	915,921	740,205	547,649	336,650	105,457	108,620	111,879	115,235	118,692	122,253
Property rights										
Members Contributions	656,861	656,861	656,861	656,861	656,861	656,861	656,861	656,861	656,861	656,861
Compulsory reserve	31,061	64,735	100,953	139,875	181,677	226,548	272,746	320,309	369,274	419,680
Profit (loss)	279,546	582,615	908,574	1,258,877	1,635,097	2,038,931	2,454,715	2,882,779	3,323,464	3,777,118
Total equity	967,468	1,304,211	1,666,388	2,055,614	2,473,635	2,922,341	3,384,322	3,859,949	4,349,599	4,853,659
Total liabilities and equity	1,883,389	2,044,416	2,214,037	2,392,265	2,579,092	3,030,961	3,496,201	3,975,184	4,468,291	4,975,912

Feasibility Indicators

There are more than one method to calculate the discounted cash flow statement. However, in this study, WACC was used to calculate the discounted cash flows and net investment value. The Company's financing structure is based on equity and loans, therefore the discounted rate should be adjusted according to WACC based on the following assumptions, the following table illustrates the net expected free cash flow of the project:

Table 38: Free Net Cash flows Table

Net Cashflows	Pre-Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027	Final Value
Net Free Cashflows	-1,642,154	385,336	481,225	496,082	511,475	527,432	543,980	561,151	578,976	597,491	616,734	9,497,886
Discount Factor	1.00	0.74	0.66	0.60	0.54	0.49	0.44	0.40	0.36	0.33	0.29	0.29
Net Present Value for Cash Flows	-1,642,154	283,643	319,834	297,696	277,133	258,033	240,290	223,808	208,498	194,275	181,062	2,788,402

Table below (40) illustrates payback period for the project, as it is one of the indicators which investors care about before taking the investment decision as they need to know when the project will return the invested amount of money. Payback period definition is the total required period so that the project generates total money equal to the total invested capital expenditure, as investor is always looking to return the full invested amount of money at the earliest possible.

Table 39: payback period

Payback Period	Before Operating	2018	2019	2020	2021	2022	2023	2024	2025	2026	2027
Net Free Cashflows	-1,642,154	385,336	481,225	496,082	511,475	527,432	543,980	561,151	578,976	597,491	616,734
Project Value	1,642,154	1,256,817	775,592	279,510	231,965	-759,397	-1,303,378	-1,864,528	2,443,504	3,040,99	3,657,73
Payback Period	3	1	1	1	0	0	0	0	0	0	0

Table 40: Financial Analysis Results

Weighted Average Cost of Capital (WACC)	%10.75
Net Present Value for Cashflows	3,630,519
Payback Period	3
Internal Rate of Return	%35.19

Sensitivity Analysis

Sensitivity analysis is conducted to investigate the effects of changes in significant inputs of the financial analysis. This includes changes in revenues, total investment costs, and operating costs.

Conclusions and Recommendations

Based on the profitability analysis and considering that project internal rate of return is higher than WACC, it is concluded that the project is feasible as the net present value for the project is positive 3,630,519 Jordan Dinars considering that the project provides 13 Job Opportunities for the governorate residents.

Table 41: Sensitivity Analysis

Sensitivity Analysis	Internal Rate of Return	Payback Period	WACC
Original Scenario	%35.19	3	10.75%
Revenues declined by 10%	%33.30	3	10.75%
Operating Expenses Increased by 10%	%20.50	6	10.75%